CLARA: Shared-Completion Certificates for Cancer-Genomics Query Clarification

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Background. Cancer-genomics frequency queries can leave patients versus specimens, assay denominators, specimen context, and mutation classes unresolved. Missing measurements introduce another uncertainty: every alternative must refer to the same unmeasured specimens. Independent identification intervals can compare incompatible worlds and falsely suggest that clarification is necessary.

Methods. CLARA enumerates 16 typed scientific specifications and represents an unprofiled specimen as absent, broad-only, or restricted-and-broad. Sharing these nested states across queries and factoring contributions by patient gives sharp absolute discrepancies and parametric relative-discrepancy bounds. Exact rational count-support comparisons decide the strict 10% relative OR 5-percentage-point (pp) cutoff. A violating-pair graph supports sufficient stopping and expected-cost or minimax clarification. A separate point guard tests whether the sharp identification interval supports a point-error budget after scientific meaning is resolved.

Results. A frozen public snapshot retains 12 TCGA cohorts, 30 genes, 6,210 specimens, and all 360 cohort–gene families. SQL and pandas counts agree on 5,760 specifications. Shared completions remove 3,048 independent-box alarms among 43,200 pairs, including 289 beyond exact equivalence merging. Uniform-intent clarification is 1.773 versus 1.833 questions for the equivalence-merged baseline (3.28%); the per-family best fixed order matches the dynamic program. After all fields are selected, 720 TCGA specifications still fail a 2.5 pp point-error budget. KIRC–VHL cohort frequency spans 32.42–53.91%, despite a fixed profiled fraction of 41.29%. A separate MSK-IMPACT cohort adds 480 verified specifications and 562 repeated-specimen patients; all selected genes occur in both source panels. Prepared-table audits of the full TCGA registry take median 1.04–1.10 seconds in serial measurements; table preparation takes 20.56 seconds separately.

Interpretation. Scientific clarification chooses a function; missing observations determine its shared input. CLARA separates semantic stability from point identification. A typed guard returns clarification, interval, or point outcomes with hash-bound, replayable provenance receipts. Forty-three software contracts, independent global TCGA optimization, 64 small-world fixtures, frozen inputs, full registries, and vector figure sources accompany the work. These are execution and mathematical checks; no language-model evaluation, researcher study, or clinical-benefit claim is made.

Disclosure. OpenAI ChatGPT (Codex) assisted throughout drafting, derivations, software, and figure design. Authors retain responsibility for validation. Sources: cBioPortal; TCGA PanCancer Atlas; Zehir et al., *Nature Medicine* (2017), doi:10.1038/nm.4333.